

Diurnal and Seasonal Variability in Carbon Monoxide at Maitri, Antarctica

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ABSTRACT

Measurement of carbon monoxide (CO) has been carried out from a pristine site, Maitri, (70.77° S, 11.43° E) Antarctica. The obtained time series data for one year period (January- December 2004) clearly indicates the occurrence of definite diurnal and seasonal variation of CO concentration in ambient air at Antarctica. Monthly mean CO concentration has generally been observed highest in the months of August (72.62 ± 24 ppbv), September (73.17 ± 28 ppbv) and October (73.25 ± 27 ppbv) and minimum in the months of January (42.77 ± 15.06 ppbv) and February (43.24 ± 14.64 ppbv). Diurnal changes were systematically observed, with higher CO concentration during daytime sunlit period. This CO enhancement in daytime sunlit period could be due to the photochemical production by photolysis of organic matter.

Keywords: - Carbon monoxide, snow firn, formaldehyde, solar insolation cycle.

INTRODUCTION

Photochemical reactions in snow have recently witnessed an unprecedented surge of interest [Couch et al., 2000, and Domine' and Shepson, 2002 and many more]. Several experiments on freshly fallen snow [Haan et al., 2001] as well as on the artificial snow [Hutterli et al., 2002] are carried out in the laboratory under controlled condition. On the other hand numerous field studies at different locations in polar region, Arctic [Ridley et al., 2000, and Sumner and Shepson, 1999] as well as Antarctic, [Davis et al., 2001, Jones et al., 2000, and Jones et al., 2001] and mid-

latitude regions [Honrath et al., 1999, Honrath et al., 2000a, Honrath et al., 2000b, Hutterli et al., 1999, Hutterli et al., 2001, and Jacobi et al., 2002] have been performed. The result obtained by Haan et al. [2001] from the field experiment and laboratory tests suggested that the organic compounds and formaldehyde photolysis could be a major substrate for the photochemical CO production in freshly fallen snow. At the same time Haan et al [2001] also observed diurnal changes in CO concentration in freshly fallen snow samples. Rammussen et al. [1982] observed large amounts of CO in snow samples collected at Mount Hood (Oregon) and attributed the abundant CO to the absorption of atmospheric CO molecules on snow crystals.

Riedel et al. [1999] investigated the annual cycle of formaldehyde at Antarctica and found that the diurnal and seasonal cycle of HCHO exists, with maximum in summer and minimum in winter. Further, they investigated that the observed concentration shows diurnal variation on clear sunny days, with maximum in early afternoon and minimum during night. Experimental results have demonstrated production of reactive nitrogen oxides in the surface layer of irradiated snow covers [Beine et al., 2002, Davis et al., 2001, Honrath et al., 1999, Honrath et al., 2000a, Jones et al., 2000 and Jones et al., 2001] through photolysis of nitrate incorporated in snow [Cotter et al., 2003 and Honrath et al., 2000b] and are subsequently released into the atmosphere. Exchange of hydrogen peroxide (H_2O_2) and formaldehyde (HCHO) between the snow and the atmosphere had been reported [Hutterli et al., 1999, Hutterli et al., 2002, Jacobi et al., 2002 and Jacobi et al., 2003]. Such an exchange of photo-chemically and chemically active gases modifies the chemical composition and the oxidation capacity of the boundary layer over the snow-covered regions [Ridley et al., 2000 and Yang et al., 2002].

In order to study the diurnal and seasonal pattern of CO, online measurements of CO were carried out at Maitri, Antarctica. Here we report the results obtained from 12 months of regular observation during 2004 period, which shows the existence of diurnal and seasonal pattern of CO in Antarctica.

Observation site and general meteorology

The observation site Maitri, Indian Antarctic Research Station (hereafter, only Maitri will be mentioned) is located on a rocky, ice-free area called Schirmacher Oasis of Queen Mand Land, Antarctica. CO analyzer was installed in small hut (Environmental laboratory) located 160 m northeast of the main station, where as the stations generator complex is located 50 m southwest of the main station. The second nearest local source of CO, Russian station's generator complex, is located at 3.5 km southeast-east of main station.

The mean annual temperature at Maitri is about -9°C , with daily averaged minimum and maximum temperature recorded was about -28.1°C and 5.5°C respectively. The maximum solar inclination angle is about 47° . The 24 hr daytime and nighttime period is generally observed between 14 November - 28 January and 20 April - 22 July respectively. As it appears from the wind rose shown in Fig. 1, surface wind directions are mostly southeast with high wind speed. This dominant wind direction along with high-speed is due to the circumpolar belt of westerly winds and the katabatic winds, that flows down the steep slopes of interior highlands and doom shape of the Antarctic continent, which is clearly evident in Fig. 1. Often wind speed was above 25 knots, crossed 90 knots occasionally with heavy snowdrift causing "blizzards". Thus, due to the prevailing wind conditions, there is no serious contamination problem over the observation site. Special efforts are made during the data analysis to filter out the data that had hints of contamination when compared with wind direction on hourly basis.

Methodology

CO measurements were carried out using gas filter correlation Non-Dispersive Infra-Red absorption (NDIR) CO analyzer of Advanced Pollution Instrument Inc (API, Model 300). The technical specifications of CO analyzer are given in Table 1. The main air inlet was installed on the top of the sidewall of the observatory 2.5 m above the ground. Teflon tube used for sampling was 2 m long with an inner diameter of 4 mm. Regular automatic zero checking every two hours was carried out and CO calibration was performed once everyday using Scotty-Martin Standard calibration gas with 1 ppmv mixture of CO in air. Out of 366 days in the year 2004, data was collected on 307 days at 5 minute interval, of which 187 days of data is used in the present analysis. On remaining 120 days, data collected was less than 60% of the entire days reading. This is due to snowdrift and 'blizzards' in Antarctica.

Table 1—Technical specification of CO analyzer used in the present experiment at Maitri, Antarctica

Zero Noise:	< 0.01 ppm RMS
Span Noise:	< 0.5% of reading above 5 ppm (RMS)
Lower Detectable Limit (LDL):	0.02 ppmv
Zero Drift:	< 10 ppb/24 hours, 20 ppb/7 days
Span Drift:	< 0.5% of reading/24 hours, 1% of reading/7 days
Linearity:	1% of full scale
Precision:	0.5% of reading
Sample Flow Rate:	1800 scc/min $\pm 20\%$

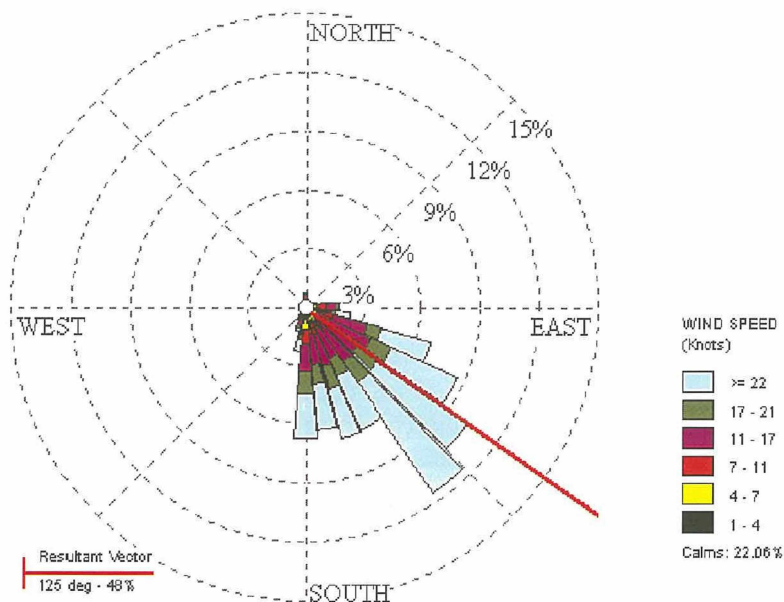


Fig. 1. Surface wind speed and wind direction at Maitri, Antarctica (based on 10 months, hourly measurement during the observation period Jan. 2004 to Oct. 2004. Hourly data of wind speed and wind direction for the month of Nov. and Dec. was not available, so six hourly data was used). Wind speed and wind direction were measured using Anemometer and wind vane respectively.

RESULTS AND DISCUSSION

We have observed definite diurnal and seasonal variation in CO concentration in surface air at Maitri. Fig. 2(a) illustrates the monthly mean distribution of CO concentration at Maitri, (during 2004) and comparison with monthly mean distribution of CO concentration at other monitoring station namely Syowa (69.00° S, 39.58° E), Palmer (64.70° S, 64.00° E), Mawson (68.11° S, 65.10° E), and Halley Bay (75.50° S, 26.60° E) stations in Antarctica. The CO concentration data is taken from World Data Center for Greenhouse Gases. The difference in the monthly mean CO concentration between Maitri and other Antarctic stations could be due to change in monitoring site and difference in sampling and measurement technique used. Frequency distribution statistics of CO concentration for observations at Maitri is given in Fig. 2(b), which shows that about 60% values lies in the first bin and approximately 20% of values lies in the second and third bin

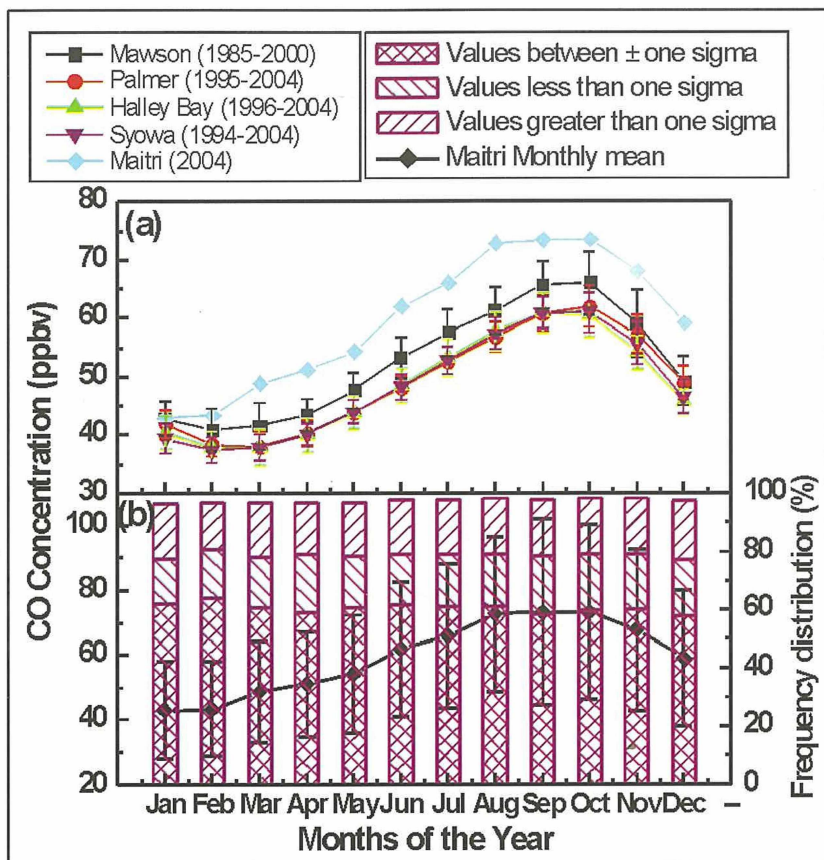


Fig. 2: (a) Comparison of averaged monthly mean CO concentration at various Antarctic stations with monthly mean CO concentration at Maitri, Antarctica for the period January to December 2004. (b) Monthly mean values of CO in ambient air (based on 48236 measurements taken at Maitri, Antarctica, for the aforementioned period), along with monthly frequency distribution (%). Vertical bar shows the one sigma standard deviation. For monthly frequency distribution (%) of CO concentration, the data set is divided into three bin (I) values between monthly mean $\pm$ sigma upper and lower bound, (II) values less than sigma lower bound and (III) values greater than sigma upper bound

each. Fig. 2(b) illustrates that the CO concentration, in a cyclic manner, gradually increases from February to September and decreases from September to February. The seasonal cycle of CO (Fig. 2(a) and (b)) is mainly driven by seasonal variation in OH concentration, which acts as a

major sink for CO, along with the seasonality of emission in Southern Hemisphere. These factors, and approximately 2 months lifetime of CO, produces a sharp decrease in polar summer and gradual increase in pre-polar and polar winter.

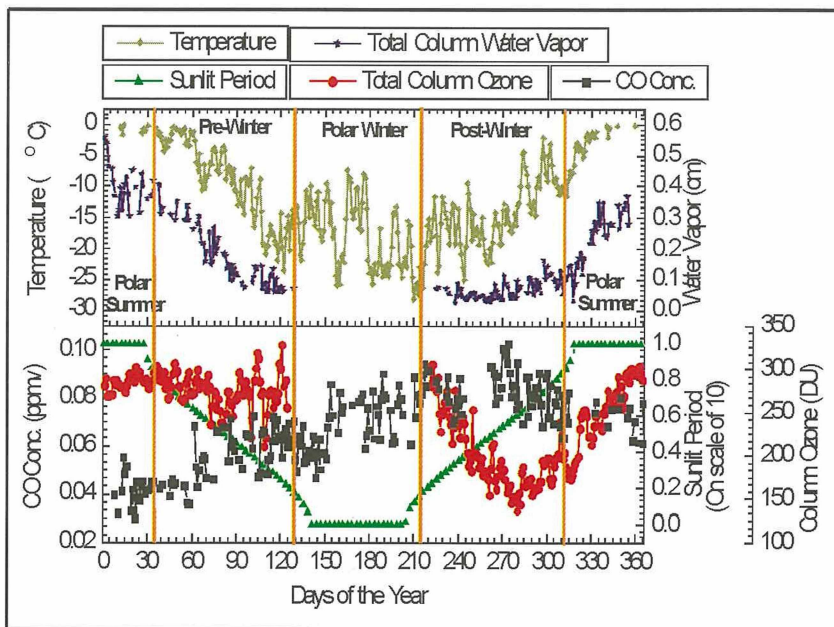


Fig. 3: Annual variation of daily averaged Temperature, Total column water vapor, CO concentration and Total column ozone along with annual variation in the sunlit period at Maitri, Antarctica.

Polar Winter (MJJ)

During polar winter, we have observed a gradual increase in CO concentration from about 54 ± 18 ppbv in May to about 65 ± 22 ppbv in July (Fig. 2(a) and (b)). During this period lot of snow furm is accumulated around the entire monitoring area. Accumulation of snow furm around Maitri is due to the frequent snowdrift that takes place during high wind speed and Antarctica blizzards, which is common in Polar Winter. It can be seen from Fig. 3 that there is non-availability of sunlight during this period. Therefore, occurrence of photochemical reactions is negligible in snow furm. It appears from Fig. 4 that the diurnal variation in CO concentration was negligible during polar winter (MJJ).

Post-Polar Winter (ASO)

During post-polar winter period, we have observed a steady state of CO concentration in the surface air from about 72 ± 24 ppbv in August to about 73 ± 27 ppbv in October (Fig. 2(a)). Till this time of the year, CO keeps on accumulating in the atmosphere and reaches to maximum value in seasonal cycle. In the first week of August, there is roughly 4 hrs of daytime, which at the end of October reaches 19 hrs of daytime. Surface air temperature increases following the course of sunlit period, causing melting of snow furn giving substantial flux of gases out of the snow pack. The high values could be due to the non availability of OH radicals, as result of less water vapor and sunlit period, and could be due to photochemical CO production. In the absence of local anthropogenic sources of CO in Antarctica, main sources of CO are photochemical oxidation of natural hydrocarbons and photochemical production of CO in snow [Haan et al., 2001].

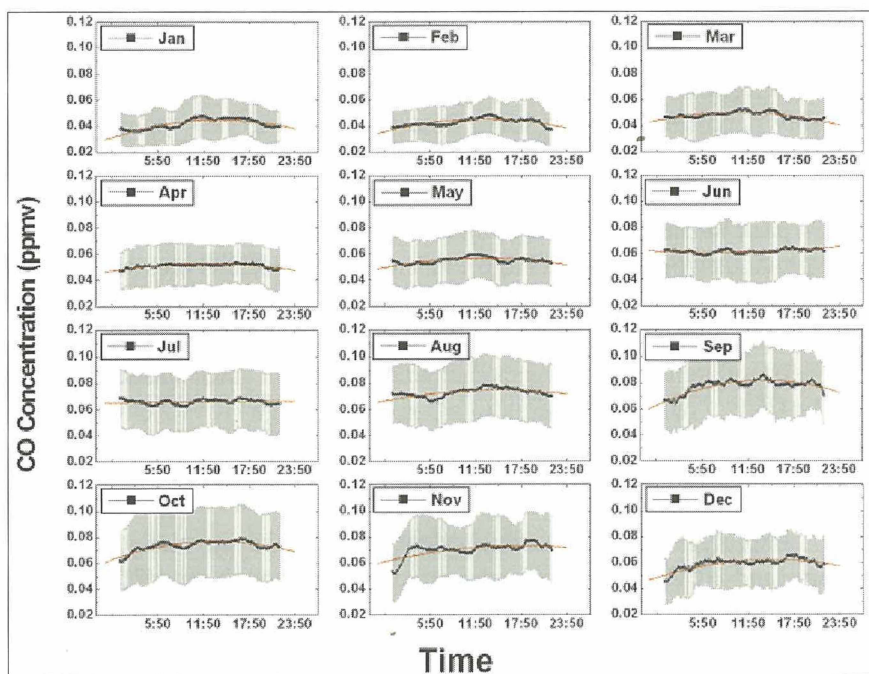
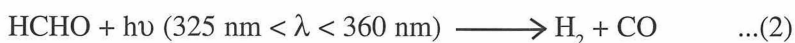
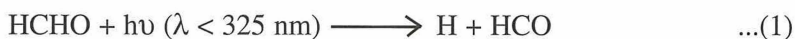


Fig. 4: Diurnal variation of average CO mixing ratio at Maitri, Antarctica for the period Jan. 2004 to Dec. 2004. All times are in local standard time (LST), which is 45 min ahead of GMT. The red lines represent the polynomial fit on each set of data.

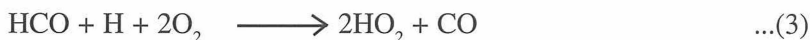
With the onset of sunlight, photochemical reactions initiate. Heterogeneous chemical reactions start in the polar stratosphere which initiates ozone depletion. This ozone depletion tends to the formation of ozone hole (Fig. 3) which leads to increase in UV irradiance at the surface. Ozone hole recover by October last or November start.

As it appears from Fig. 4, in the month of August, a prominent diurnal cycle existed due to daytime build up of CO concentration. The amplitude of diurnal cycle of CO concentration increases and attains maximum value in the month of September with the increasing hours of daytime. This diurnal cycle could be due to the photolysis of organic matter during daytime sunlit period and loss to the snow during nighttime. CO production by photolysis of HCHO [Haan *et al.*, 2001] produced through buried winter snow furn layer [Sumner and Shepson 1999], along with oxidation of CH₄ [Riedel *et al.*, 1999] plays a vital role in the diurnal cycle of CO in Antarctica. Measurements of snow and gas phase HCHO in the Arctic by Sumner and Shepson [1999], suggested photochemical formaldehyde production at the air-snow interface, resulting in the substantial flux out of the snow-pack. At the same time, Hutterli *et al.* [1999] also confirmed de-gassing process for HCHO from winter snow furn layer at Greenland. HCHO concentration in freshly deposited snow at South Pole was found to be 6 ppbw, of which 25% gets quickly released into the atmosphere. At the same time concentration of the order of 6 ppbw was found in the snow just below the snowpack surface, reflecting high concentration of HCHO preserved in the previous winter's snow. High concentration of HCHO present just below freshly deposited snow, in the previously accumulated snow surface apparently results in temperature dependent air snow exchange. Understanding this air-snow exchange is important since surface snow is a large HCHO reservoir leading to a potentially large HCHO emission from the snowpack in post polar winter and polar summer, altering the local boundary layer photochemistry.

Photolysis of HCHO, which, depending on the wavelength, yields H atoms and HCO radicals or hydrogen and CO through reaction (1) and (2) [Riedel *et al.*, 1999],



The products of reaction (1) react rapidly with the molecular oxygen to form hydroperoxyl radicals (HO_2) and carbon monoxide through reaction (3) [Sumner and Shepson, 1999],



By the end of September, and in the month of October, the daytime build up of CO was less prominent. This is due to the cumulative effect of (I) decrease in the extent of snow burn, decreasing the production of CO through photolysis of HCHO and (II) increase in the concentration of OH radicals due to increase in the level of water vapor and sunlit period, results in the oxidation of CO through reactions (4) and (5),



As a result of this, the amplitude of diurnal cycle of CO concentration reduces in the month of October as it can be seen in Fig. 4.

Polar Summer (NDJ)

During polar summer period, we have observed sharp decline of surface concentration of CO from around 67 ± 25 ppbv in November to 42 ± 15 ppbv in January (Fig. 2(a)). Polar summer begins around 14 November and lasts till 28 January with 24 hrs of sunlit period. By the mid of November, polar vortex breaks and ozone hole recovers completely. As it appears from Fig. 3, surface air temperatures also attend its maximum value during this period. By first week of the November almost all of the snow burn, previously deposited during polar winter and post polar winter, melts and even the fresh snow deposited during this period, due to snowdrift, melts in no time. The OH concentration is very high compared to other time of the year due to increase in the water vapor level. The combined effect of this, as it appears from Fig. 4, is the existence of a comparatively weaker but considerable diurnal variation during this period.

Pre-Polar Winter (FMA)

In Pre-polar winter period, we have observed gradual increase in surface CO concentration from about 43 ± 14 ppbv in February to 51 ± 16 ppbv in April (Fig. 2(a)). At Maitri, pre-polar winter period starts from February onwards till April. As it appear from Fig. 3, surface air temperature

and water vapor decreases along with sunlit period, resulting in the reduction of OH concentration. At the same time fresh snow furm starts to accumulate, with prolonged melting period initially and by the mid of pre-polar winter melting of snow furm eventually stops. The temperature dependent air-snow exchange of HCHO and related photochemical reactions goes on reducing and then stops. Due to this the diurnal variation was weak, when compared with post-polar winter and polar summer diurnal variation, as it appears in Fig. 4.

CONCLUSIONS

The results obtained from the first year-round time series online field measurements of CO concentration at 5 minute interval from a high latitude site (Maitri, Antarctica) shows the occurrence of systematic diurnal cycle. The diurnal cycle of CO implies that photochemical production of CO is active in the snow covered region of Antarctica and confirms that relatively strong local sources of CO exist and are activated during sunlit period. We expect that, organic matter trapped in snow and especially photochemical destruction of formaldehyde (HCHO) is probably the major substrate of the photochemical formation of CO when exposed to sunlight. *Sumner and Shepson* (1999) and *Riedel et al* (1999) showed diurnal cycle of HCHO in the surface air and showed that HCHO photolysis depending on the amount of actinic flux yields CO and can be a significant source in the lower polar troposphere. In conclusion, further studies of diurnal and seasonal pattern of CO in polar region are required for the better estimate of the global CO budget.

ACKNOWLEDGEMENT

We thank to the Director, NPL and Head, RASD, for their encouragement during the progress of the work. Thanks are also due to the NCAOR (MoES) and CSIR for logistic and financial assistance. We are also thankful to Leader and members of 23rd ISEA. We are greatly thankful to Indian Metrological Department, www.meteo.infospace.ru, World Data Center for Greenhouse Gases (<http://gaw.kishou.go.jp/wdcgg.html>), and U.S. Naval Observatory, Astronomical Application, for meteorological data, carbon monoxide data and sunrise – sunset data respectively.

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